TITLE PAGE

Problem Statement ID - Q1

Problem Statement Title - Eco-Loop Building Agents

PS Category - Software

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Autonomous closed-loop building energy control:

EnergyPlus digital twin x open-source LLM x MCP

IDEA TITLE

Proposed Solution

A two-tier controller wrapped around a live EnergyPlus simulation. The LLM does not sit in the control loop - it writes the control policy the loop executes.

Tier 2 - Cognitive (once per simulated day)

Open-source LLM reads building state through MCP tools, then emits a setpoint policy as validated JSON

Tier 1 - Reflex (every timestep)

Pure arithmetic, no I/O, cannot raise. Applies the policy, clamps to a hard safety envelope, writes setpoints via the EnergyPlus API

Why this design

An annual run is 35,040 timesteps - an LLM call per timestep is both too slow and one malformed reply away from a dead run. Decoupled, the simulation completes even if the model server dies mid-run.

Innovation

Supervisory MPC with an LLM policy layer; setpoint safety enforced twice over (schema bounds, then an independent clamp at the actuator); setback driven by measured occupancy, not a clock

Measured, 7-day July A/B, LLM-authored policy: -6.76% electricity -7.64% peak demand PMV -0.391 to +0.14

TECHNICAL APPROACH

Technologies

Python 3.11 | EnergyPlus 26.1 Python API (pyenergyplus) | Model Context Protocol (FastMCP) | Ollama - Llama 3.2 | Pydantic | Streamlit + Plotly | pytest

Closed-loop data flow

EnergyPlus -> SensorBus (handle binding, warmup + design-day filtering, rolling aggregation)

-> MCP tool surface -> LLM -> Policy JSON -> Pydantic validation

-> PolicyExecutor (hard clamp) -> set_actuator_value -> back into EnergyPlus

Tool-calling architecture

7 MCP tools, implemented once and exposed twice - to the in-process agent and to any external MCP client, so behaviour cannot drift

commit_policy is the only writer. It returns structured rejections (offending field + reason) which are fed back into the retry prompt, so the model corrects one field instead of resampling blindly

Latency management

Agent runs on a daemon thread with a busy guard and a 20 s timeout; one invocation per simulated day means 365 LLM calls per annual run, not 35,040

Handling lengthy simulation logs

Severity filter, numeric normalisation, then deduplicate with counts - a .err file of tens of thousands of lines collapses to a handful of messages. Telemetry is aggregated to min/mean/max; the tools cannot return raw arrays

FEASIBILITY AND VIABILITY

Feasibility - already working, not a proposal

Running end to end on the DOE reference small office (5 conditioned zones, Chicago TMY3). 7-day baseline vs controlled A/B complete. 41 unit tests green.

Pure software: no hardware, no proprietary licences, no cloud dependency - the LLM runs locally.

Challenges met during build, and how

LLM emits an out-of-range setpoint -> Pydantic bounds reject it, feedback drives a corrected retry; an independent clamp at the actuator makes an unsafe value unreachable

LLM server dies mid-run -> reflex tier continues on the last-known-good policy and the simulation still completes

Actuator or meter name absent from the model -> probed from eplusout.edd and the API data dump; binding fails loudly instead of writing nothing

Simulation silently ran design days only -> filtered on kind_of_sim, and the simulated span is now reported every run

Comfort scored while the building was empty -> PMV and unmet hours now measured only against measured occupancy

Viability and impact

Buildings are ~40% of global energy use. A 5% cut needs no capital works - it is a supervisory layer over the existing BMS, and the same loop retargets to any EnergyPlus model by editing one config file.

ARTIFACTS

Measured results - 7-day July A/B

Policy authored by the LLM (llama3.2:3b, local, temperature 0 so the run is reproducible). Identical model and weather; only the controller differs.

Electricity 1251.97 -> 1167.33 kWh -6.76%

Total energy 1309.46 -> 1225.14 kWh -6.44%

Peak demand 18.97 -> 17.52 kW -7.64%

Comfort - occupied hours only

Mean PMV -0.391 -> +0.140 band -0.5 to +0.5

PMV in band 82.8% -> 97.3% +14.5 pp

Unmet setpoint 0.00% -> 0.60% of occupied zone-hours

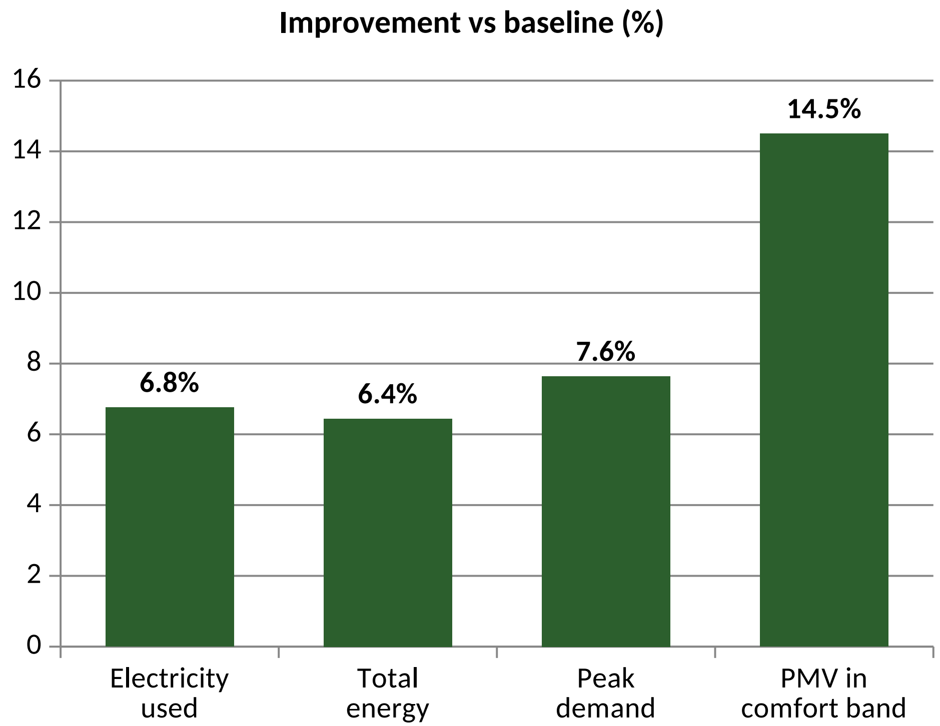
The stock schedule overcools (PMV -0.39), so raising the occupied cooling setpoint saves energy AND moves comfort toward neutral - aligned, not in tension. The agent beat a hand-tuned static policy (-5.08%) on the same model.

Artifacts

Source, both .idf models + runtime policy snapshots, savings dashboard, architecture document, agent trace (tool calls, rejections, self-corrections)

Repository: github.com/DivyanshM30/eco-loop-building-agents

PoC video (3 min): drive.google.com/drive/folders/1BnIvuYQWBIXOHk3vVQ0LGkk2cF6vWLvq



RESEARCH AND REFERENCES

EnergyPlus - Using EnergyPlus as a Library (Runtime and Data Exchange API)
bigladdersoftware.com/epx/docs/23-2/input-output-reference/api-usage.html

EnergyPlus Python API reference
energyplus.readthedocs.io/en/latest/runtime.html

EnergyPlus API examples - NREL/EnergyPlus repository
github.com/NREL/EnergyPlus/tree/develop/tst/EnergyPlus/api

DOE Commercial Reference Buildings - Small Office, New Construction 2004
energy.gov/eere/buildings/commercial-reference-buildings

Model Context Protocol specification
modelcontextprotocol.io

ASHRAE Standard 55 / Fanger PMV-PPD thermal comfort model
EnergyPlus Engineering Reference, Thermal Comfort chapter

Ollama - local open-source model serving
ollama.com